
Stock Trading Strategy using Multivariate Time Series Forecasting

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Abstract

Multivariate Time Series Forecasting (MTSF) is an important application of autoregressive models that spans several key fields. We focus on the stock price prediction aspect of these models and aim to build an application on top of an efficient MTSF model. In MTSF, it is important to capture long-range dependencies in data, to which end we test multiple architectures and styles of forecasting models to select the lightest and most efficient model for stock price forecasting. We aim to make an accessible application that can effectively predict intra-day stock prices at different horizons.

1 Introduction

Our problem statement is “How can we efficiently inform a user whether to buy, sell, or hold stocks based on a multihorizon stock price prediction?”. This problem is composed of two parts: Creating a lightweight stock prediction model and creating an effective stock purchasing strategy using the predictions from this model. To complete the first task, we explore various MTSF models, which are detailed in Methods. The trading strategy built on top of this is based on several existing frameworks and combines three ideas in multi-horizon forecasting, transaction cost aware execution, and differential weighting. Further details on this strategy are detailed in Methods.

2 Related Work

2.1 Stock Price Prediction Using Triple Barrier Labeling and Raw OHLCV Data: Evidence from Korean Markets [4]

This paper trains Deep Learning models on raw OHLCV (Open, High, Low, Close, Volume) for stock price forecasting. Previous work referenced in this paper makes use of engineered features to predict stocks, which is shown to be unnecessary by the authors. They apply a simple LSTM model to raw OHLCV data and achieve similar performance to models trained on engineered features. This approach inspired us to focus on using RNN-based frameworks as well as raw OHLCV data instead of engineered data.

2.2 A Comprehensive Survey of Deep Learning for Multivariate Time Series Forecasting: A Channel Strategy Perspective[9]

This paper offers a structured survey of deep learning approaches for multivariate time series forecasting (MTSF), with a unique focus on modeling correlations among channels (i.e., variables). It proposes a three-level taxonomy for channel strategies: channel independence (CI), channel dependence (CD), and channel partiality (CP). The survey analyzes mechanisms (Transformer, CNN, MLP, GNN) and correlation characteristics, namely asymmetry, lagginess, polarity, group-wise behavior,

dynamism and multiscale effects. It compares strengths, weaknesses, and implementation costs of each strategy, then outlines future research directions for foundation models and multimodal forecasting. This paper gave us an excellent understanding of the possibilities for model selection and techniques.

2.3 The Statistical Recurrent Unit[8]

This work presents the Statistical Recurrent Unit (SRU), an un-gated recurrent neural architecture that substitutes complex gates in LSTMs/GRUs with moving averages of learned summary statistics. SRUs retain long-term sequence dependencies by maintaining statistics at multiple scales, offering many viewpoints of recent and distant past data through simple linear operations. Empirical results show that SRUs often outperform LSTMs and GRUs, especially on tasks with profound long-term dependencies. SRUs capability to learn and maintain multi-scale dependencies may enhance temporal pattern recognition in stock data, especially for trading strategies sensitive to historical context.

2.4 Trading Strategy

Our strategy synthesizes three established frameworks: multi-horizon forecasting, transaction cost aware execution, and differential weighting. Regarding the former, Zhang and Zohren[11] demonstrated that encoder decoder models that leverage multiple forecast horizons outperform single-horizon approaches, specifically at longer time scales. This result is particularly valuable to our project, as predicting multiple minute horizons from minute OHLCV data is considered longer term relative to high frequency trading strategies. Additionally, Wen et al.[10] showed that multi-horizon quartile forecasts improve decision-making under uncertainty by capturing different temporal dynamics simultaneously. This motivated using four LSTM horizon outputs rather than just one.

The differential weighting (horizon bias) derives from Garleanu and Pederson’s[3] finding that predictors with faster mean reversion should dominate portfolio construction. Their “aim portfolio” framework explicitly weights signals by persistence. Specifically, since one minute predictions decay fastest, they receive the highest weight for immediate execution decisions. Mukherjee et al.[7]) also validated performance-based weighting for intraday trading, confirming that recency and relevance improve signal quality.

To simulate aspects of real market conditions, our strategy also accounts for transaction cost by implementing a threshold-based execution mechanism. Luu and Song[6] proved that optimal trading policies under fixed transaction costs require threshold-based decision schemes, where trades are only executed when profit exceeds friction. Specifically, we assume that the cost of a transaction does not exceed 0.3% of its order’s value. We chose this threshold value based on works such as Aked and Masturzo[1] and Frazzini et al.[2], which show that transaction costs are generally in the range of 0.3% to 0.75%.

3 Methods

3.1 Prediction model

We decomposed the MSTF into three parts: a long term trend for the data that spans over decades, seasonal trend which represents weeks, months, or individual years, and a residual to capture day-to-day fluctuations.

$$Y(t) = L(t) + S(t) + R(t)$$

$$L(t) = \beta_2 e^t + \beta_1 t + \beta_0$$

$$S(t) = \sum_{i=1}^n \alpha_i \sin(nt) + \gamma_i \cos(nt)$$

$L(t)$ is represented using an exponential plus a linear term to model the general trend of the graph

$S(t)$ is decomposed into series of sine and cosine curves (essentially a fourier transform)

Finally the $R(t)$ which is the residual term and will be modeled using a recurrent neural network taking input from all of the MSTF features.

To train the model specifically, we can convert the linear time space into a feature space that includes the exponential term for $L(t)$ and the sinusoidal terms for $S(t)$. Then, we can perform a linear regression in that feature space. To train $R(t)$, we set up a true $R(t) = Y(t) - (L(t) + S(t))$ to conduct supervised learning using LSTM layers followed by a couple of dense layers.

3.2 Trading Strategy

3.2.1 Multi-Horizon Signal Construction

Let P_t denote the closing price at time t , and let $\hat{P}_{t+h}$ represent the model's predicted price at horizon $h \in \{1, 10, 30, 60\}$ minutes ahead, where predictions are made at the current time $t = 0$. We define the expected return for each horizon as:

$$r_h = \frac{\hat{P}_h - P_0}{P_0}, \quad h \in \{1, 10, 30, 60\}. \quad (1)$$

We then construct a composite signal S as a weighted linear combination of the predicted returns:

$$S = w_1 r_1 + w_{10} r_{10} + w_{30} r_{30} + w_{60} r_{60}, \quad (2)$$

where the weights must satisfy $\sum_h w_h = 1$ and will initially be set to $w_1 = 0.5$, $w_{10} = 0.3$, $w_{30} = 0.15$, and $w_{60} = 0.05$.

The weighting scheme reflects the empirical considerations that shorter-horizon forecasts exhibit provide more immediate actionable signals[3], and longer-horizon forecasts inform directional consistency and filter short-term noise. The decreasing weight structure balances immediacy against trend confirmation, ensuring that the strategy is not overly reactive to high-frequency fluctuations.

3.2.2 Transaction-Cost-Aware Thresholding

To prevent excessive trading in possible real world transaction costs, we implement a dynamic execution threshold τ that scales with forecast uncertainty. Trading occurs only when the composite signal exceeds this threshold in magnitude, ensuring that the expected edge justifies transaction costs. The threshold is defined as:

$$\tau = \max \left(0.003, \frac{2 \cdot \text{RMSE}}{P_0} \right), \quad (3)$$

where RMSE is the root mean squared error of the model's 1-minute-ahead predictions on the validation set, and P_0 is the current price. The lower bound of 0.003 (0.3%) represents a conservative estimate of transaction costs for large-cap equities[2][1]. The uncertainty-scaled component $2 \cdot \text{RMSE}/P_0$ ensures that the threshold expands when the model exhibits higher prediction variance, imposing stricter execution criteria during periods of uncertainty.

3.2.3 Position Logic and Execution Rules

At each timestep, we define a state $\pi_t \in \{\text{LONG}, \text{SHORT}, \text{HOLD}\}$ based on the composite signal S and threshold τ :

$$\pi_t = \begin{cases} \text{LONG}, & \text{if } S > \tau, \\ \text{SHORT}, & \text{if } S < -\tau, \\ \text{HOLD}, & \text{if } |S| \leq \tau. \end{cases} \quad (4)$$

The HOLD state represents either no position or continuation of the previous position when the signal remains inside the neutral zone $[-\tau, \tau]$. This prevents rapid position flipping due to small signal perturbations under the threshold boundary.

Position exits and reversals follow three rules:

1. **Reversal:** If the current position is LONG and $S < -\tau$, exit the long position and enter a short position (and vice versa).
2. **Flattening:** If an active position exists and $|S| \leq \tau$, close the position and return to HOLD.
3. **End-of-day liquidation:** Close all positions 5 minutes before market close.

4 Preliminary Results

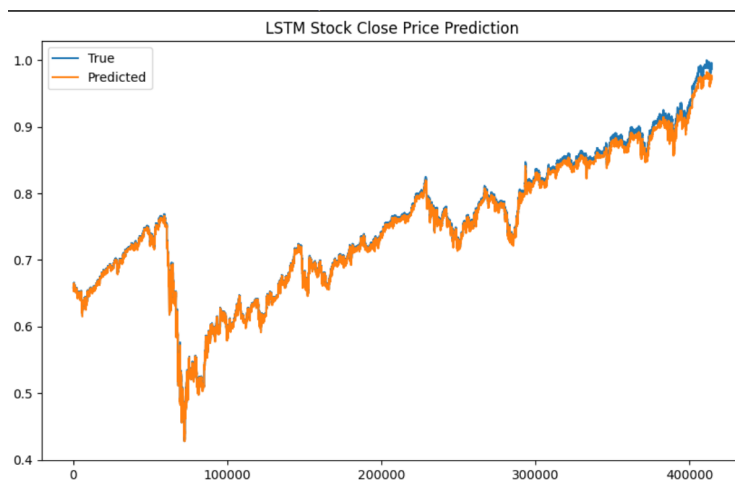


Figure 1: Stock price prediction using LSTM

The image above is a comparison between the actual data and the predicted data of a LSTM neural network for one-minute forecasts.

Table 1: R-squared values for different parameters

Parameter	R-squared Value
Open	0.9629946295834333
High	0.9630288183591993
Low	0.9629604075550279
Close	0.9629929946249869
Volume	0.20897027938784984
BarCount	0.15448741283673006
Average	0.9629949144060919

The linear regression models $L(t) + S(t)$ for the major stock price predictors have an R-squared value are at least 0.96 showing the significance of long-term and seasonal trends in the stock data.

5 Future Work and Work Allocation

We plan to train our new model using the LSTM and linear components, then implement our trading strategy on its multi-horizon predictions (all group members, early Nov.). We also plan to test other model architectures, such as Temporal Convolutional Networks [5] and the Statistical Recurrent Unit, while considering transformer-based models [8] (Aditya and Vibhas, mid Nov.). Each of these models will be tested on stock exchange data from various countries to ensure they are not skewed towards any specific data. We will be withholding a few years worth of data to test the models on. We will also improve the trading strategy based on empirical results (Jefrey and Rish, mid Nov.).

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